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OFFLINE NETWORK RESILIENCE FIELD MANUAL

Community Communications When the Grid Goes Down

First Edition — 2026

"When infrastructure fails, community survives."

Designed for printing and USB distribution. No internet connection required.

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SECTION 1

FOUNDATIONAL PRINCIPLES

1.1 Why Offline Networks Matter

Modern society has built its communications infrastructure on a foundation that is simultaneously powerful and fragile. The internet, cellular networks, and telephone systems all depend on centralized physical infrastructure — data centers, fiber optic cables, cellular towers, and power grids — that can fail suddenly and completely.

When that infrastructure fails, communities that have done nothing to prepare are left silent.

The vulnerabilities are well-documented. A single large-scale power grid failure can simultaneously disable every internet router, cellular base station, and landline switching center across hundreds of square miles. Natural disasters sever fiber lines and topple towers. Cyberattacks on internet service providers (ISPs) can deny entire regions access to digital communications. In moments of mass emergency — exactly when communication matters most — conventional systems become overwhelmed or non-functional.

The case for redundancy is simple: no critical system should have a single point of failure. Commercial aviation, nuclear power, and military communications all rely on layered, redundant systems for this reason. Community communications must adopt the same philosophy. A community that has invested in offline-capable mesh networks, licensed radio operators, and physical information distribution will be able to coordinate rescue, distribute resources, and maintain social cohesion when all other systems have gone dark.

Layered communications means maintaining multiple independent pathways for information to travel. If the mesh network node is down, use radio. If radio is jammed or operators are unavailable, use physical runners. If runners cannot travel, use pre-positioned bulletin boards and printed directories. Each layer compensates for failures in the others.

The **"no single point of failure" principle** demands that the community network be designed so that the loss of any one component — one node, one operator, one power source — does not silence the whole system. Distributed design, redundant nodes, trained operators, and pre-positioned materials all contribute to this goal.

1.2 The Three Pillars of Resilience

This manual organizes community communications resilience around three complementary pillars. Together, they form a layered system that is far more robust than any single technology or approach.

Pillar 1 — Local Mesh Networks

A **mesh network** is a system of devices that communicate directly with each other without requiring a central server or internet connection. Each device in the mesh

both sends and receives messages, and also *relays* messages for other devices in the network. This means messages can travel across the network by hopping from node to node even when no direct line of sight or central infrastructure exists. Mesh networks based on LoRa (Long Range) radio technology, running software such as **Meshtastic**, can cover kilometers per node and are resilient to individual node failures — if one node goes down, the network automatically routes around it.

Pillar 2 — Radio Communications

Radio is the oldest and most battle-tested emergency communications technology. It does not require the internet, cell towers, or any centralized infrastructure. Licensed amateur (ham) radio operators can communicate locally on VHF/UHF frequencies, regionally on HF (shortwave) frequencies, and even globally under the right conditions. Unlicensed services such as FRS (Family Radio Service) provide simple, accessible short-range communication for households without any training. Radio communication is immediate, real-time, and works as long as the transmitter has power.

Pillar 3 — Physical Information Distribution

When electronics fail or power is unavailable, **physical information distribution** becomes the backbone of community communication. This pillar encompasses trained information runners who carry written messages on foot or bicycle, community bulletin board networks at known fixed locations, pre-printed directories and reference materials, and dead drop locations for asynchronous message exchange. Physical distribution requires no technology, no power, and no infrastructure — only people, paper, and pre-agreed protocols.

These three pillars are not competitors; they are complements. Each fills gaps in the others. A community that has developed all three has true communications resilience.

1.3 Legal and Ethical Framework

FCC Licensing

Radio communications in the United States are regulated by the **Federal Communications Commission (FCC)**. Different radio services have different licensing requirements:

- **FRS (Family Radio Service):** No license required. Low-power handheld radios. Fixed channels, limited range. Legal for all users.
- **GMRS (General Mobile Radio Service):** Requires an FCC GMRS license ($35, 10-year term, no exam). Covers the licensee and immediate family. Higher power than FRS, repeater-capable.
- **Amateur (Ham) Radio:** Requires passing an FCC-administered exam for a Technician, General, or Extra class license. No fee for the license itself (exam fee varies by VEC). Provides access to a wide range of frequencies and modes.
- **CB Radio:** No license required. 40 channels in the 27 MHz band. Limited range and often congested.

Operating on licensed frequencies without proper authorization is a federal violation. Community members should obtain appropriate licenses before an emergency, not during one.

Legal Use of Mesh Networking Tools

Software mesh networking tools such as Meshtastic operate on ISM (Industrial, Scientific, and Medical) bands — specifically 915 MHz in the United States — that are unlicensed for low-power use. These tools are legal to use without a radio license for standard messaging. Mesh networking software on smartphones using WiFi operates on licensed spectrum managed through device certification; no user license is required. Always verify that the hardware and frequency band you use comply with current FCC Part 15 rules.

Privacy Considerations

Community mesh networks are shared infrastructure. All participants should understand that messages may be visible to other network participants unless encrypted. **Do not transmit sensitive personal information** — medical details, financial information, home security status — over unencrypted channels. The network should have clear policies about data retention, who can access message logs, and how user information is protected.

Community Consent and Governance

The community communications network is a shared resource and must be governed by the community it serves. No individual or organization should control the network unilaterally. Governance decisions — about channel encryption keys, node placement, access policies — should be made transparently, with community input, and documented in writing.

Mutual Aid Ethics

The guiding ethical principle of this network is **mutual aid**: the network exists to serve the whole community, not to benefit those who built it. In practice, this means: open access for all community members, no gatekeeping based on technical skill, prioritizing the most vulnerable (elderly, disabled, non-English-speaking residents) in communications and resource allocation, and transparent operation at all times.

1.4 Threat Assessment Matrix

Threat	Likelihood	Impact	Primary Comms Failure	Recommended Response
Power Grid Failure	High	Critical	Internet, cell towers, landlines all fail within hours as backup generators exhaust fuel	Activate mesh network; switch to battery/solar-powered radios; deploy runners; open bulletin boards
Internet Backbone Disruption	Medium	High	Internet-dependent services (VOIP, streaming, apps) fail; cell may remain via SMS	Activate local mesh network; use radio for coordination; switch to offline-stored data
Natural Disaster (Tornado/Flood)	Medium–High	Critical	Physical infrastructure destroyed; roads blocked; cell towers damaged; power out	All three pillars activated simultaneously; runners navigate on foot; radio for resource coordination
Cyberattack	Low–Medium	High	Internet	Activate local

Threat	Likelihood	Impact	Primary Comms Failure	Recommended Response
on ISPs			severed at provider level; local WiFi may still function; cell may be unaffected	mesh network immediately; maintain radio readiness; activate USB data kits
Cell Tower Overload	High	Moderate	Voice calls fail; SMS may still work; data throughput collapses	Avoid cellular for routine comms; use mesh network and radio; preserve cell for true emergencies
Physical Infrastructure Damage	Medium	High	Fiber cuts, tower collapses sever specific areas; outage may be localized	Route communications around damaged area via radio; alert mutual aid partners; document damage

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SECTION 2

MESH NETWORKING SETUP

2.1 What Is a Mesh Network?

In a conventional home network, all devices connect to a single router, which connects to the internet through your ISP. The router is the central hub. Remove it, and no device can communicate with any other. This is the fragile model that most people rely on every day.

A **mesh network** works differently. In a mesh, every device can connect directly to every other device within radio range. There is no central router. When Device A wants to send a message to Device D, but they are out of direct range, Device B and Device C — which are in range of both — automatically relay the message. The

network **self-heals**: if Device C fails, the network finds another path. No single device is critical to the whole system.

For community emergency communications, the most practical mesh technology is **LoRa (Long Range)** radio, running firmware such as Meshtastic. LoRa uses low-power radio signals in the 915 MHz band (US) to transmit text messages across distances of 1–10 km (urban to rural line-of-sight). Nodes require minimal power — a small solar panel and battery are sufficient for continuous operation. The hardware is inexpensive and widely available.

A mesh network is not a replacement for the internet. It carries text messages, GPS position data, and small data packets. It is a neighborhood-scale communication system that works entirely without external infrastructure.

KEY CONCEPT

In a mesh network, every node is both a client (sending and receiving messages) and a router (relaying messages for other nodes). There is no center. This is what makes it resilient.

2.2 Recommended Software Platforms

Platform	Type	Range	Setup Difficulty	Cost	Best Use Case
Meshtastic (LoRa radio)	LoRa radio firmware + app	1–10 km per hop	Low–Medium	$30–$60/ node	Primary community mesh backbone; text messaging, GPS position sharing
goTenna Mesh	LoRa radio + smartphone	1–6 km per hop	Low	$179/pair	User-friendly option for non-technical participants; pairs with smartphone

Platform	Type	Range	Setup Difficulty	Cost	Best Use Case
Briar	Android app (WiFi/Bluetooth)	~100 m (Bluetooth), ~50 m (WiFi Direct)	Very Low	Free	Short-range messaging between smartphones in same area; no hardware required
Meshtastic + Raspberry Pi	Store-and-forward gateway node	Extended (gateway function)	Medium–High	$80–$150/node	Fixed community hub node with message storage, BBS function, gateway capability
BATMAN-adv (Linux)	WiFi mesh protocol (kernel module)	Up to ~300 m per hop (WiFi)	High	Hardware cost only	High-bandwidth local mesh; requires Linux-capable hardware; suitable for fixed sites
Reticulum Network Stack	Cryptographic networking layer	Varies by transport layer	High	Free (open source)	Secure, encrypted communications over any medium; advanced users; extensible platform

RECOMMENDATION

Meshtastic on LoRa hardware is the recommended primary platform

for most communities. It has the best balance of range, cost, ease of use, and active development. Use Briar as a supplementary short-range option for smartphone users.

2.3 Hardware Requirements

A basic community mesh node requires the following components:

- **LoRa radio module:** A Meshtastic-compatible board such as the TTGO T-Beam v1.2 (with GPS) or Heltec LoRa 32 v3. These boards combine a microcontroller, LoRa radio module, and (in the case of the T-Beam) GPS receiver on a single circuit board.
- **Antenna:** The stock whip antenna (included with most boards) is adequate for basic nodes. For improved range, a J-pole or slim-jim antenna made from 300-ohm twin-lead provides gain over the whip. For directional point-to-point links, a Yagi antenna provides significant gain but must be aimed precisely.
- **Single-Board Computer (gateway/hub nodes):** A Raspberry Pi 4 or Zero 2W running Meshtastic firmware or a Python-based integration enables store-and-forward messaging, bulletin board service (BBS), and MQTT bridging for future internet integration.
- **Power system:** A 20W–40W solar panel, a 12V LiFePO4 battery (20–40 Ah), and an MPPT charge controller. A properly sized system will run a LoRa node 24/7 even through several cloudy days.
- **Weatherproof enclosure:** An IP65-rated outdoor electrical enclosure protects electronics from rain and humidity. Route antenna cable through a weatherproof SMA bulkhead connector.

Basic Node Parts List

Item	Approximate Cost (USD)	Purpose
TTGO T-Beam v1.2 (LoRa + GPS board)	$35–$45	Primary node radio/processor
915 MHz J-Pole or Slim-Jim Antenna	$15–$30	Improved omnidirectional range
SMA pigtail cable (coax, 1–3 ft)	$8–$15	Connects board to antenna
20W Solar Panel (12V)	$25–$50	Primary power generation
LiFePO4 Battery 12V 20Ah	$60–$100	Energy storage for nights/cloudy days

Item	Approximate Cost (USD)	Purpose
MPPT Charge Controller (10A)	$20–$40	Manages solar charging safely
5V USB Step-Down Converter	$8–$15	Powers board from 12V battery
IP65 Waterproof Enclosure	$15–$30	Weatherproof housing
SMA Bulkhead Connector + Cable Gland	$8–$12	Weatherproof antenna passthrough
Mounting hardware (mast clamp, U-bolts)	$10–$20	Secures enclosure and antenna
TOTAL ESTIMATED COST	**$204–$357**	Per fully solar-powered node

2.4 Network Topology Planning

A **star topology** places all nodes connected to one central hub. This is simple but creates a single point of failure: if the hub fails, the network collapses. A **chain topology** links nodes in a line — useful for corridor coverage (a road or river valley) but vulnerable to any break in the chain. A **mesh topology** connects nodes to multiple neighbors, creating multiple paths for messages to travel. The mesh is the recommended approach for community emergency communications.

Node Placement Strategy

- **Height is paramount:** LoRa radio range is dramatically improved by elevation. Nodes mounted on rooftops, water towers, church steeples, or hills will cover far more area than ground-level nodes.
- **Line-of-sight:** LoRa performs best with clear line-of-sight between nodes. Urban environments with dense buildings will reduce range to 1–3 km per hop; rural or elevated environments may achieve 10+ km.
- **Redundant coverage:** Each area of the community should be reachable through at least two independent nodes. Plan node placement so no single node loss isolates any neighborhood.
- **Coverage radius planning:** Assume 2–3 km radius per urban node and 5–10 km per elevated rural node for planning purposes. Test actual coverage after installation.

Channel Planning and Naming Conventions

- Use a single community channel with a pre-shared encryption key for all nodes.
- Consider a separate unencrypted "public" channel for newcomers to connect during emergencies.
- Name nodes clearly: [NEIGHBORHOOD]-[TYPE]-[NUMBER] (e.g., NORTHSIDE-ROOF-01, DOWNTOWN-GATEWAY-01).
- Document all node names, locations, and custodians in the Node Registry (Section 9.2).

2.5 Step-by-Step Node Deployment

1. **Flash Meshtastic firmware to the board.** Download the correct firmware file for your specific board model from the official Meshtastic website. Use the Meshtastic Web Flasher or ESPHome Flasher to write the firmware via USB. Verify the flash was successful by connecting to the device via the Meshtastic app.
2. **Configure device name, region, and channel.** Using the Meshtastic app (Android or iOS) via Bluetooth, set the device name per community naming convention, set the region to US, configure the community channel name, and enter the pre-shared encryption key (PSK) provided by the Network Coordinator.
3. **Set GPS coordinates.** If the board has a GPS module (T-Beam), allow it to acquire a satellite fix before mounting. If no GPS is equipped, manually enter the node's coordinates using the fixed position feature in the app settings.
4. **Test radio-to-radio link before mounting.** Bring the configured node within range of an existing network node. Send a test message from the new node and verify receipt on the network. Verify the new node appears in the device list on neighboring nodes.
5. **Mount antenna at highest practical point.** Install the antenna at the highest accessible point on the structure. Ensure the antenna is vertical

(omnidirectional) unless using a directional Yagi. Secure all cable connections and weatherproof antenna connections with self-amalgamating tape.

6. **Connect power source; verify solar charging.** Connect the solar panel to the MPPT charge controller, the battery to the controller, and the step-down converter output to the board's USB port. Verify the charge controller shows a charging state and the board powers on. Check battery voltage.
7. **Log node in the community node registry.** Record the node ID, physical location, custodian name and contact, hardware model, installation date, and GPS coordinates in the Node Registry (Section 9.2). Distribute updated registry to all Network Coordinators.
8. **Perform range test with a handheld unit.** Walk the expected coverage area with a handheld Meshtastic device, sending messages at intervals. Note signal strength readings. Document coverage boundaries and any dead zones. Update the community coverage map.

2.6 Network Security Basics

Channel encryption: Meshtastic supports AES-256 encryption on a per-channel basis. Every community node and participant device must use the same pre-shared key (PSK) to participate in an encrypted channel. Generate a strong, random PSK (use the Meshtastic app's built-in key generator) and distribute it securely to all authorized participants in person or via encrypted message — never over unencrypted radio or the internet.

Pre-shared key management: Treat the community PSK like a physical key to a building. Keep a written record of the PSK in a secure location (the Network Coordinator's go-box, a sealed envelope at the community emergency meeting point). Change the PSK if it is believed to have been compromised. Never transmit the PSK over the mesh network itself.

Operational Security (OPSEC): Even with encryption, practice good OPSEC on the mesh network:

- Do not transmit home addresses, medical conditions, financial information, or personal vulnerabilities.

- Do not discuss the precise locations of stored supplies or valuable equipment in detail.
- Be aware that metadata (who is transmitting, GPS position, time) may be visible to other network participants even on encrypted channels.
- Use first names or call signs rather than full identifying information in messages.

Physical security of nodes: Fixed community nodes represent shared infrastructure and should be protected. Mount nodes out of easy reach where possible. Do not mark enclosures with conspicuous labeling. The Network Custodian responsible for each node should periodically verify the node has not been tampered with.

SECTION 3

RADIO COMMUNICATIONS

3.1 Radio Frequency Overview

Radio waves are part of the electromagnetic spectrum. Different frequency ranges have different propagation characteristics that make them suitable for different communication purposes. Understanding these basics allows the community communications team to select the right tool for the right situation.

- **HF (High Frequency, 3–30 MHz):** HF radio waves can travel thousands of miles by reflecting off the ionosphere — a layer of charged particles in the upper atmosphere. This makes HF the basis for long-distance and even intercontinental communications. Requires an Amateur General or Extra class license for most HF bands. Equipment is larger and more complex. Best for connecting with regional emergency nets, coordinating with distant mutual aid partners, or receiving news from outside an affected area.
- **VHF (Very High Frequency, 30–300 MHz):** VHF propagates primarily by line-of-sight. Range is typically 5–50 km depending on antenna height and

terrain. The 2-meter band (144–148 MHz) is the most popular amateur VHF band and the backbone of most local emergency communications. NOAA Weather Radio also broadcasts on VHF. Requires Technician class license or above for amateur use.

- **UHF (Ultra High Frequency, 300–3000 MHz):** UHF offers better penetration through buildings than VHF, making it advantageous in dense urban environments. The 70-centimeter band (420–450 MHz) is popular for local amateur communications and linked repeater systems. FRS and GMRS radios operate in the UHF band (462–467 MHz).

3.2 License Classes and Requirements

License Type	Frequency Access	Range	Exam Required	Cost	Best For
FRS (no license)	462.5625–467.7125 MHz (22 channels)	0.5–3 km	None	Free	Household and neighborhood short-range communication
GMRS (FCC license)	462–467 MHz; higher power; repeater use	1–50 km (with repeater)	None (fee only)	$35 / 10 years	Family and neighborhood networks; repeater-based community nets
Amateur Technician	VHF/UHF amateur bands; some HF privileges	Local to regional (via repeater)	35-question exam	~$15 exam fee	Local emergency nets; APRS; mesh gateway operation
Amateur General	All Technician + most HF bands	Regional to international	35-question exam (after Technician)	~$15 exam fee	HF emergency comms; Winlink; long-distance coordination
Amateur Extra	All amateur bands, all	Global	50-question exam (after	~$15 exam fee	Full emergency

License Type	Frequency Access	Range	Exam Required	Cost	Best For
	privileges		General)		capability; serving as net control; EMCOMM leadership

ACTION ITEM

Every community emergency communications team should have a minimum of

three licensed Amateur Technician operators

and at least one

Amateur General or Extra class operator

for HF capability. Encourage all interested community members to study for and obtain their Technician license — it is achievable in one weekend of study.

3.3 Recommended Radio Equipment

Tier 1 — Basic (Under $50)

Entry-level equipment accessible to all households. Every household in the community network should have at minimum a Tier 1 option.

- **Baofeng UV-5R or UV-82:** Dual-band (VHF/UHF) handheld radio. Inexpensive and widely available. Requires programming (use CHIRP software). Note: transmitting on amateur frequencies requires a Technician license; the UV-5R is legal to use for receive-only or on FRS/GMRS channels within legal power limits.
- **FRS Bubble-Pack Radios:** Simple 22-channel FRS radios (Motorola T100, Cobra FRS, etc.). No programming needed. Legal for all users. Good for household and neighbor-to-neighbor communication.

Tier 2 — Intermediate ($50–$300)

Suitable for dedicated community radio operators and neighborhood node operators.

- **Yaesu FT-60R:** Rugged, reliable dual-band handheld. Excellent receiver sensitivity. Recommended for serious emergency communicators.
- **GMRS Mobile Radio (Midland MXT400 or similar):** 40W GMRS mobile radio for vehicle or base station use. Extends range significantly. Requires GMRS license.
- **VHF/UHF Dual-Band Handheld with Programming Cable:** Any quality dual-band handheld (Kenwood TH-D74A, Yaesu FT-3DR) with APRS and digital mode capability.

Tier 3 — Advanced ($300+)

For community hub stations, go-boxes, and operators providing HF/regional coverage.

- **Icom IC-7300:** HF/50 MHz transceiver. Direct-sampling SDR architecture provides excellent performance. 100W output. Ideal for community HF base station. ~$1,000–$1,200.
- **Yaesu FT-991A:** All-band, all-mode transceiver covering HF through 70 cm. Good choice for a single versatile community station. ~$900–$1,100.
- **RTL-SDR Receiver:** A $25–$35 software-defined radio dongle for receiving a wide range of signals without transmitting. Excellent for monitoring weather, APRS, and other channels. No license required for receive-only.
- **Go-Box Kit:** A complete, portable station in a weatherproof case (see Appendix D). Includes transceiver, power supply, antennas, cables, and operating supplies. Deployable in under 15 minutes.

3.4 Emergency Frequencies

Frequency	Band	Use	License Required	Notes
162.400–162.550 MHz	VHF	NOAA Weather Radio — 7 channels, continuous broadcast	None (receive only)	All weather-capable radios should monitor this. Critical for storm warnings.
146.520 MHz	2m VHF Amateur	National 2-meter Simplex Calling Frequency	Amateur Technician+	Monitor this when activating. Make contact, then move to a working frequency.
462.675 MHz	UHF GMRS	GMRS Channel 20 — nationally recognized GMRS calling/emergency	GMRS License	Recommended for community GMRS nets. Monitor continuously during activations.
27.065 MHz	HF CB	CB Radio Channel 9 — national emergency/distress channel	None	Use only for genuine distress if no other means available.
462.5625 MHz	UHF FRS/GMRS	FRS Channel 1 — common starting point for FRS users	None (FRS power)	Useful for coordinating with households using basic bubble-pack radios.
144.390 MHz	2m VHF Amateur	APRS National Frequency (North America)	Amateur Technician+	Position and text messaging via APRS digipeaters. Monitor for position data.
Local Repeater: _____	_____	Local community repeater — fill in with your community's assigned repeater	Varies	Primary repeater: _____ / Offset: _____ / Tone: _____
915 MHz (LoRa)	UHF ISM	Meshtastic default channel (US region)	None (Part 15)	Primary mesh network data channel. Text messages, GPS position, telemetry.

Frequency	Band	Use	License Required	Notes
3.985 MHz / 7.285 MHz	HF	National Traffic System (NTS) net frequencies	Amateur General+	For formal message handling to/from outside the affected area.
14.300 MHz	HF	Maritime Mobile Service Network — worldwide emergency	Amateur General+	International emergency coordination; monitor if seeking outside help.

3.5 Operating Procedures

NATO Phonetic Alphabet

Use the NATO Phonetic Alphabet (see Appendix A) to spell out names, call signs, and critical information clearly over radio. Always spell out call signs, addresses, and any ambiguous words phonetically.

Standard Radio Procedure

9. **Listen before transmitting.** Always listen for several seconds before pressing the transmit button (PTT — Push To Talk) to confirm the channel is clear.
10. **Identify yourself.** Begin every transmission with your call sign (or community identifier if unlicensed): "This is [CALL SIGN], at [LOCATION], with traffic for [RECIPIENT]."
11. **Keep transmissions brief.** Say what you need to say concisely. Long transmissions block the channel, drain your battery, and are harder to receive clearly.
12. **Pause for replies.** After transmitting, release PTT and wait for acknowledgment before sending more information.
13. **Confirm receipt.** When you receive a message, confirm with "COPY" or "ROGER" and your call sign. If you did not receive clearly, say "SAY AGAIN" or "SAY AGAIN ALL AFTER [last word heard]."

14. **Sign off properly.** End your transmission with your call sign: "This is [CALL SIGN], clear."

The SALUTE Report Format

Use the SALUTE format to report emergency information clearly and completely (see Appendix B for full detail):

- **S** — Size: How many people / vehicles / items involved
- **A** — Activity: What is happening
- **L** — Location: Where (address, landmark, grid coordinates)
- **U** — Unit/Uniform: Who is involved (civilians, responders, etc.)
- **T** — Time: When was this observed
- **E** — Equipment: What resources are present or needed

Net Control Procedures

A **net** is a scheduled, organized radio session. Net Control is the designated operator who manages the net. Net Control calls the net to order, takes check-ins from participants, manages traffic (messages to be passed), and closes the net. Detailed net control procedures and a sample script are provided in Section 8.3.

3.6 Repeater Networks and Linking

A **repeater** is an automated radio station, typically located at an elevated site, that receives signals on one frequency (the input) and simultaneously retransmits them at higher power on another frequency (the output). This dramatically extends the effective range of handheld and mobile radios. A community repeater can allow a handheld radio with a 1-mile ground-level range to communicate across 30+ miles.

Find local amateur repeaters using the RepeaterBook directory (available offline in USB kit). For GMRS repeaters, consult MyGMRS.com before the emergency and record frequency/tone information in Section 9.3.

APRS (Automatic Packet Reporting System) is a digital system operating on 144.390 MHz that transmits GPS positions, weather data, and short messages. APRS digipeaters relay packets across a region. An APRS-capable radio (Kenwood TH-

D74A, Yaesu FT3DR) or a smartphone with APRSDroid can participate when the internet is unavailable.

Local Repeater Directory (Fill In)

Repeater Name	Output Freq.	Input Freq./Offset	CTCSS Tone	Mode	Notes

3.7 Digital Radio Modes

Digital voice modes offer improved audio quality and spectrum efficiency compared to analog FM:

- **DMR (Digital Mobile Radio):** Used by many GMRS and amateur repeater systems. Allows two simultaneous conversations on one channel via time-division. Requires a DMR-capable radio (Anytone, TYT, etc.).
- **D-STAR (Digital Smart Technologies for Amateur Radio):** Icom's digital voice and data protocol. Supports data transmission alongside voice. Requires D-STAR capable radio and repeater.
- **System Fusion (Yaesu):** Yaesu's C4FM digital mode. Automatic analog/digital switching. Wires-X linking system connects repeaters globally via internet (internet-dependent).

Digital data modes allow transmission of text and files over radio frequencies:

- **JS8Call:** A weak-signal HF mode derived from FT8. Designed for messaging and store-and-forward networking over HF. Can relay messages through other stations. Excellent for community HF data networks.
- **Winlink:** An email-over-radio system. Uses Pactor, VARA, or packet radio protocols to send and receive email without internet access. Widely used in

emergency communications for formal message traffic and resource requests.

- **VARA:** A high-performance HF digital mode used with Winlink. Requires a soundcard interface between radio and computer. Significantly higher throughput than older Pactor modes.
- **PSK31:** A simple keyboard-to-keyboard HF digital mode. Very narrow bandwidth. Good for one-on-one HF text communication.
- **APRS:** Packet-based position reporting and short messaging on 144.390 MHz. Self-sufficient digital network independent of internet.

SECTION 4

PHYSICAL INFORMATION DISTRIBUTION

4.1 The Information Runner System

The **Information Runner System** is the community's last-resort communication backbone — and, paradoxically, one of its most reliable. It requires nothing more than willing people and paper. Runners are designated community members who physically carry written messages between nodes when all electronic communications have failed. This system has been used effectively in every major disaster where conventional communications failed, from post-hurricane neighborhoods to urban blackouts.

Runner Routes

Pre-define runner routes that connect all neighborhood nodes, bulletin board locations, community hubs, and designated meeting points. Routes should be walked and mapped in advance. Routes should have alternatives in case of road blockage or hazard. Route maps are stored in the USB kit and as laminated printed cards carried by each runner.

Message Priority System

- **Priority 1 — URGENT:** Immediate threat to life; medical emergency; fire; rescue needed. Runners drop all other traffic to deliver Priority 1 messages immediately.
- **Priority 2 — HIGH:** Resource requests; welfare checks on vulnerable residents; critical supply needs.
- **Priority 3 — ROUTINE:** Status updates; general information; community coordination messages.

Handoff Procedures

Every message carried by a runner must include: date/time written, originating location, destination, priority level, message text, and sender's name. The runner logs the message on their Runner Log Sheet (template in USB kit). Upon delivery, the recipient signs and dates the log. The runner retains the log as a record. A copy of every message is retained at the originating location.

Runner Safety

Runners should never travel alone at night or in hazardous conditions. Assign runners in pairs. Runners should carry a whistle, a flashlight, a copy of their route map, and a Meshtastic handheld radio if available. Runners are not required to cross actively dangerous areas — they should report the hazard instead.

4.2 Community Bulletin Board Network

A network of physical bulletin boards at fixed, known locations provides a passive information distribution system that functions 24/7 without power or personnel. Community members check bulletin boards for updates during an emergency, just as one might check a town square notice board.

Location Selection

Bulletin boards should be located at places where people naturally congregate or pass:

- Community centers and recreation centers
- Public libraries

- Fire stations and police stations
- Churches, mosques, synagogues, and temples
- Schools (exterior facing, accessible when closed)
- Neighborhood association offices
- Laundromats, pharmacies, and other essential businesses

Standardized Board Format

Each board should have designated zones: **EMERGENCY ALERTS** (top, red border), **RESOURCE AVAILABILITY** (left, green border), **NEEDS AND REQUESTS** (right, yellow border), and **GENERAL INFORMATION** (bottom, white). Use waterproof backing and a protective cover where possible. Include the date of last update prominently.

Posting Protocols

- Any community member may post to the NEEDS and GENERAL zones.
- The EMERGENCY ALERTS zone is managed by the community Network Coordinator or designated bulletin board custodian only.
- All posts must include: date, time, name of poster, and a contact method if possible.
- Outdated posts (older than 24 hours during an active emergency) should be moved or removed by the board custodian.
- Unverified information should be clearly marked "UNCONFIRMED — DO NOT ACT WITHOUT VERIFICATION."

4.3 Printed Information Systems

The community must pre-position printed materials *before* an emergency occurs. Printed materials serve communities when every screen is dark and every battery is dead. Priority printed materials include:

- **Emergency Contact Directory:** Names, addresses, call signs, and secondary contacts for all community roles. Updated regularly and distributed to all households.

- **Neighborhood Map with Node Locations:** A printed map showing all mesh network nodes, bulletin board locations, community meeting points, and runner routes.
- **This Field Manual:** Every household that participates in the network should have a printed copy. Print on both sides, saddle-staple, laminate the cover.
- **Quick-Cards:** Laminated cards (3x5 inch) with critical reference information — emergency frequencies, phonetic alphabet, SALUTE format, signal codes. See Section 9 for reference tables suitable for quick-card printing.
- **Resource Lists:** Pre-printed lists of community resources: where to find water, first aid, shelter, food, power for charging medical equipment.

Print quality and weather resistance: Print on 24 lb or heavier paper. Laminate critical reference cards and cover pages. Store printed materials in zip-lock bags or waterproof document folders. Rotate printed materials on the same schedule as digital data updates (see Section 6.5).

4.4 Dead Drop Locations

A **dead drop** is a pre-agreed location where written messages can be left and collected without the sender and receiver needing to make direct contact. Dead drops allow asynchronous communication when direct contact is impossible, dangerous, or inadvisable. The concept is simple: I know where to leave a message for you; you know to check that location at agreed intervals.

Establishing Dead Drops

- Select locations that are fixed, durable, and unlikely to be disturbed: a specific flagstone, a hollow tree, behind a loose brick, inside a weatherproof container attached to an agreed structure.
- Each dead drop location should be known only to the parties who use it plus the Network Coordinator (in a sealed record).
- Assign each dead drop a code name (not a description of the location): e.g., "CEDAR," "KEYSTONE," "PILGRIM."

- Document all dead drop locations in the sealed Network Coordinator emergency records (not in any publicly distributed document).

Message Format and Authentication

Messages left at dead drops should include: a pre-agreed header code confirming the sender, date and time, priority level, and message content. To authenticate messages, use a pre-agreed code word or phrase that is changed periodically. Messages without the correct authentication code should be treated as unverified.

Maintenance Schedule

Dead drop locations should be checked weekly during normal operations (by the designated custodian) and every 6–12 hours during an active emergency. Check for weather damage, accessibility, and security at each visit.

4.5 Visual Signaling

Visual signals provide instant status information across distances that would require a runner or radio in the absence of a direct view. Standardized visual signals allow community members to communicate basic status at a glance.

Color Flag / Tape System

Color	Signal Meaning	Application Method
GREEN	All Clear / Occupants Safe / Resources Available	Green flag, green tape, or green cloth on door/window/mailbox
YELLOW	Caution / Assistance Needed But Not Urgent / Entry with Care	Yellow flag or tape. May accompany a paper note with details.
RED	Help Needed Urgently / Medical Emergency / Distress	Red flag, red tape, red cloth — most visible location on structure
BLACK	Do Not Enter / Structural Hazard / No Survivors	Black tape in X pattern on door. Used by trained Search and Rescue only.
BLUE	Medical Resources / First Aid Available Here	Blue cross symbol on window or door
WHITE	Information / Message Left Here / Check Bulletin Board	White paper or cloth with arrow pointing to message location

Window Signs

Pre-print window signs (8.5 x 11 inch, visible from the street) with the following messages. Laminate and store in each household emergency kit: "ALL SAFE — CHECK IN COMPLETE," "NEED WATER," "NEED MEDICAL HELP," "EVACUATED — CHECK [LOCATION]," "HAVE EXTRA FOOD/WATER TO SHARE."

Chalk Marking System

During search and rescue operations, use standard SAR chalk markings: an X on a door indicates the structure has been searched. Numbers inside the X quadrants indicate: top = date/time of search, left = searching team ID, right = hazards found, bottom = number of survivors removed / bodies found. Only trained SAR personnel should make or interpret door markings.

SECTION 5

POWER AND ENERGY RESILIENCE

5.1 Power Hierarchy for Communications

Communications equipment is only as reliable as its power source. A comprehensive power strategy uses multiple power sources in a defined priority order, transitioning smoothly from one to the next as each is depleted or becomes unavailable.

15. **AC Mains (Grid Power):** Normal operating condition. All equipment runs on utility power. Maintain battery backups charged at all times.
16. **Battery Backup (UPS):** A UPS (Uninterruptible Power Supply) provides immediate, seamless transition when grid power fails. Size your UPS to power critical communications equipment (router, radios, charging stations) for a minimum of 4 hours. Larger battery banks can extend this to 24+ hours.
17. **Generator:** A generator can power equipment for as long as fuel lasts. Connect via a transfer switch — never directly to household wiring. Reserve

generator use for critical equipment only to conserve fuel. See Section 5.4 for safety requirements.

18. **Solar Power:** Solar panels with a battery bank provide indefinite power in clear weather and extended operation through nights and cloudy periods with adequate battery sizing. The recommended primary power source for fixed mesh nodes and the go-box. See Section 5.2.
19. **Human Power (Hand Crank):** Last resort. Hand-crank generators and hand-crank radios provide minimal power for emergency communication when all other sources are exhausted. See Section 5.5.

Sizing backup power: Determine the power consumption (in watts) of all critical communications equipment. Common loads: Meshtastic node (0.3–1W), handheld radio on receive (0.5W), handheld on transmit (2–5W), HF transceiver receive (50W), HF transceiver transmit (100–200W), laptop (30–60W). Total the loads and multiply by hours of operation to get watt-hours (Wh) required.

5.2 Solar Power Basics

Panel sizing: A solar panel's watt rating reflects its output under ideal (full sun, perpendicular angle) conditions. In practice, expect 4–5 "peak sun hours" per day in most US locations. A 40W panel produces approximately 160–200 Wh/day. A community mesh node drawing 1W continuously needs 24 Wh/day — a 20W panel is more than sufficient. A go-box with an HF radio operating 2 hours/day at receive may need 150+ Wh/day — a 60–100W panel array.

Battery bank sizing: Size the battery bank to provide 2–3 days of operation without solar input (for cloudy weather autonomy). Multiply daily Wh consumption by 2–3, then divide by battery voltage and depth-of-discharge (50% for lead-acid, 80% for LiFePO4). Example: 100 Wh/day × 3 days ÷ 12V ÷ 0.8 = 31 Ah minimum LiFePO4 battery.

Charge controllers: Always use a charge controller between the solar panel and battery. **PWM (Pulse Width Modulation)** controllers are inexpensive and reliable for small, simple systems. **MPPT (Maximum Power Point Tracking)** controllers are 15–30% more efficient and are recommended for systems larger than 100W or when panel and battery voltages differ.

Simple Sizing Worksheet

Parameter	Value	Units
Total equipment power consumption		Watts (W)
Daily operating hours		Hours/day
Daily energy consumption (W × hours)		Watt-hours (Wh/day)
Desired autonomy without sun		Days
Total battery storage needed		Wh
Battery capacity at 12V (÷12, ÷0.8 for LiFePO4)		Amp-hours (Ah)
Peak sun hours per day (use 4 for conservative estimate)	4	Hours
Minimum panel wattage needed (Wh/day ÷ 4)		Watts (W)

5.3 Battery Systems

Lead-Acid Batteries

The oldest rechargeable battery technology. Available in flooded (maintenance-required), AGM, and gel versions. Advantages: inexpensive, widely available, well-understood. Disadvantages: heavy, must not be discharged below 50%, shorter lifespan (300–500 cycles), sensitive to temperature extremes. Store upright in a ventilated area (flooded cells off-gas hydrogen).

LiFePO4 (Lithium Iron Phosphate)

The recommended battery chemistry for emergency communications. Advantages: very safe (no thermal runaway risk unlike other lithium chemistries), longer lifespan (2,000–5,000 cycles), can discharge to 80% depth without damage, much lighter than lead-acid, operates in wider temperature range. Disadvantages: higher initial cost (2–4× lead-acid per Wh). **Recommended for all new battery bank installations.**

Warning Signs of Failing Batteries

- Battery does not reach full charge voltage despite full charging period
- Voltage drops rapidly under light load
- Battery case is swollen, bulging, or leaking
- Battery becomes very hot during charging
- Unusual smell (sulfur from lead-acid, chemical smell from lithium)

> **WARNING**
>
> Never connect different battery types (lead-acid and lithium) in the same bank. Never charge lithium batteries with a charger designed for lead-acid without confirming the charger supports LiFePO4 chemistry. Improper charging can cause fire or explosion.

5.4 Generator Safety

Carbon monoxide (CO) poisoning is the leading cause of death from generator use during disasters. CO is colorless, odorless, and lethal. Never operate a generator indoors, in a garage, in a basement, or near any open window or door. Operate generators at least 20 feet (6 meters) from any structure opening. Install battery-powered CO detectors in your home.

Fuel storage: Gasoline degrades in as little as 30 days without a fuel stabilizer additive (use Sta-Bil or equivalent). Store gasoline in approved containers, in a cool, dry, ventilated area away from living spaces and ignition sources. Propane stores indefinitely and is the preferred fuel for long-term storage. Practice fuel rotation: use stored fuel regularly and replace it.

Transfer switch: Never connect a generator directly to household wiring without a proper transfer switch. Back-feeding power into utility lines can electrocute utility workers restoring power. A manual transfer switch or interlock kit installed by a licensed electrician is essential for any generator connected to household circuits.

Fuel rotation schedule: Gasoline — rotate every 3 months, treat with stabilizer. Propane — inspect tanks annually for rust and valve condition. Diesel — rotate every 12 months, treat with biocide additive to prevent algae growth.

5.5 Human-Powered Options

- **Hand-crank radios:** Emergency radios (Midland ER310, Eton FRX5-BT, etc.) combine AM/FM/NOAA weather radio with a hand-crank generator and USB charging port. Adequate for monitoring; not sufficient for sustained transmitting.
- **Bicycle generators:** A bicycle connected to a permanent magnet alternator via friction drive or hub attachment can generate 30–100W of continuous power with moderate pedaling effort. Sufficient to charge batteries, power a radio, or charge phones. Inexpensive DIY builds are possible with a car alternator and a bicycle frame.
- **Manual Morse code:** A battery-powered or crystal-controlled Morse code transmitter requires only a small battery that can be hand-charged. Experienced CW (Morse code) operators can communicate over thousands of miles with minimal power — some QRP (low power) CW setups operate on milliwatts.
- **Solar chargers for smartphones:** A 10W foldable solar panel can maintain a smartphone charge through daylight hours. Pair with a power bank (10,000+ mAh) to extend charging through the night. The phone is still useful for offline apps and as a Meshtastic Bluetooth terminal even without cellular service.

5.6 Power Conservation

When power is limited, conservation allows communications to continue far longer. Apply these practices immediately when grid power fails:

- **Duty cycle radios:** Rather than transmitting continuously, establish scheduled "traffic windows" — specific times when the net is active. Between windows, radios stay on receive or are powered down.

- **Low-power modes for mesh nodes:** Meshtastic nodes can be configured to increase their sleep interval, reducing power consumption from ~100mA to under 10mA during inter-message periods.
- **Screen brightness:** Reduce display brightness on all devices to the minimum usable level. A phone screen is one of its highest power draws.
- **Prioritize power allocation:** Establish a written power priority list. Suggested order: (1) mesh network nodes, (2) primary radio station receive, (3) lighting for coordination activities, (4) medical equipment, (5) radio transmit, (6) device charging.
- **LED lighting:** Replace all temporary lighting with LED lanterns and flashlights. An LED lantern draws 1–3W versus 50–100W for an incandescent equivalent.
- **Consolidate charging:** Designate a single community charging station rather than charging devices individually at multiple locations. This improves efficiency and allows better power management.

SECTION 6

DATA AND INFORMATION MANAGEMENT

6.1 Offline Data Storage Strategy

The internet is not just a communications medium — it is the primary repository of information that modern people rely on every day. When it fails, access to medical references, maps, contact directories, technical manuals, and government resources disappears simultaneously with communications. Communities that have stored critical data locally — on USB drives, SD cards, and laptop hard drives — retain access to the information they need to function.

The principle is simple: store it before you need it. Download offline maps, reference materials, community directories, and critical software *during normal times*, update them regularly, and distribute them widely enough that the loss of any single storage device does not cause irreplaceable data loss.

What to store:

- **Maps:** OpenStreetMap offline data for your region and surrounding areas. Neighborhood-scale maps with node locations, runner routes, and community resources marked.
- **Medical references:** Offline first aid guides, medication reference guides, and emergency medical procedures.
- **Community contact directories:** Names, addresses, call signs, and emergency contacts for all community members and key roles.
- **Government resources:** FEMA guides, local emergency management plans, utility company emergency procedures.
- **Technical documentation:** This field manual, Meshtastic documentation, radio programming files, equipment manuals.
- **Offline encyclopedias:** Wikipedia offline via Kiwix provides a comprehensive reference spanning medicine, history, science, and technical subjects — all without internet.

6.2 Recommended Offline Apps

App	Platform	Use	Approximate Size	Notes
Meshtastic	Android / iOS	Mesh network client — text messaging, GPS, node management	~30 MB	Requires Meshtastic-compatible LoRa hardware via Bluetooth. Core app.
Briar	Android	Encrypted messaging via Bluetooth, WiFi Direct, or Tor	~15 MB	Works without internet. Contacts must be added in person via QR

App	Platform	Use	Approximate Size	Notes
				code first.
Maps.me	Android / iOS	Offline maps with navigation and search	Varies by region (50–500 MB)	Download region maps before emergency. Works without cell or WiFi.
OsmAnd	Android / iOS	Advanced offline mapping with topographic, hiking, and nautical layers	Varies by region	More features than Maps.me; steeper learning curve. Preferred for technical users.
Kiwix	Android / iOS / Windows / Linux	Offline Wikipedia and other reference content from ZIM files	App ~15 MB; ZIM files up to 80+ GB	Pre-download the English Wikipedia ZIM (~85 GB full, ~6 GB simplified). Store on USB.
First Aid (American Red Cross)	Android / iOS	Offline first aid procedures and emergency guidance	~50 MB	Download content while online for full offline access. Include with USB kit APKs.
HAM Study	Android / iOS	Amateur radio license exam preparation	~20 MB	Pre-emergency use for licensing. Offline capable once downloaded.
APRSDroid	Android	APRS client for Android via audio or TNC	~10 MB	Works offline for local APRS operation. Requires APRS-capable radio connection.
CHIRP	Windows / Mac / Linux	Radio programming software for Baofeng and many other radios	~15 MB	Pre-program all community radios with channels before emergency. Store CHIRP files on USB.

6.3 USB Kit Contents

The **Community USB Kit** is a standardized USB flash drive (minimum 32 GB) containing all data, software, documentation, and resources needed for community emergency communications. Multiple copies should be distributed to the Network Coordinator, each Node Custodian, radio operators, and the community supply cache. Update all kits simultaneously on the scheduled update day.

Folder 1 — MANUALS

- This field manual (PDF, current edition)
- Amateur radio license study guides (Technician, General)
- Meshtastic official documentation and user guide
- ARRL Emergency Communications Handbook (if authorized)
- Equipment-specific manuals for all community radios and hardware
- FEMA IS-100, IS-200, IS-700 course materials

Folder 2 — MAPS

- Offline map tiles for the full region (OsmAnd or Maps.me export)
- Neighborhood PDF maps with node locations, runner routes, bulletin boards, and community resources
- Regional topographic maps (USGS 7.5-minute series)
- Building floor plans for key community structures (community center, library, etc.)

Folder 3 — SOFTWARE

- Meshtastic firmware files (all supported board variants)
- Meshtastic Web Flasher (offline capable version)
- Briar APK (Android installer)
- Kiwix app (Windows, Android, Linux installers)
- Wikipedia ZIM file (simplified English, minimum)
- CHIRP radio programming software and pre-configured channel files

- VeraCrypt (for encrypted storage)
- 7-Zip (for archive management)

Folder 4 — MEDIA

- Training videos (MP4 format, no internet required): Meshtastic setup, radio programming, solar power basics
- Quick-card PDFs (emergency frequencies, phonetic alphabet, SALUTE format, signal codes)
- Workshop slide decks from community training sessions

Folder 5 — DATA

- Community contact directory (VeraCrypt encrypted container — password managed by Network Coordinator)
- Community resource lists (food, water, medical, power)
- Emergency frequency list (current, with local repeater information)
- Local APRS digipeater list
- Radio channel programming files (.img or .csv format)

Folder 6 — TEMPLATES

- Message forms (runner system, formal radio traffic)
- Runner Log Sheets
- Node Registry forms
- Incident Log templates
- After-Action Review templates
- Exercise planning templates

6.4 Data Security and Privacy

Encryption for USB drives: The community contact directory and any sensitive personal information on USB kits must be encrypted. Use **VeraCrypt** to create an encrypted container on each USB drive. VeraCrypt is free, open-source, and well-

audited. The password for the encrypted container must be memorized by the Network Coordinator and shared with one designated backup — not written on or near the USB drive.

Archive encryption: For individual encrypted files or folders, 7-Zip's built-in AES-256 encryption is adequate. Use long, complex passwords (20+ characters).

Password management without internet: Keep a written record of critical passwords in a sealed, signed envelope stored in a secure, known location (a safe, a fireproof box). The envelope should be dated and updated whenever passwords change. Access should require two people — no single person should be the sole keeper of critical passwords.

What NOT to store on community USBs: Social Security numbers, financial account information, full medical records, detailed personal security information (alarm codes, safe combinations), information about domestic violence situations, and any information that could endanger individuals if the USB is lost or stolen.

6.5 Information Freshness

Offline data has a shelf life. Contact directories change. Software is updated. Maps improve. Procedures evolve. A structured update schedule ensures that community data remains current and reliable.

- **Update schedule:** The first Sunday of each month is the designated Community Data Update Day. The Supply Custodian coordinates updates to all USB kits, printed materials, and bulletin board directories.
- **Version numbering:** Each edition of community documents uses the format YYYY.MM (e.g., 2026.05 for May 2026). Never distribute a document without a version number and date.
- **Change logs:** Each updated document includes a change log listing what was added, changed, or removed since the previous version. The USB kit includes a CHANGELOG.txt at the root level listing all updates to all files in the kit.
- **Responsibility:** The Training Coordinator is responsible for updating documentation. The Supply Custodian is responsible for physical distribution.

The Network Coordinator signs off on all changes to sensitive data (contact directory, encryption keys).

SECTION 7

COMMUNITY ORGANIZATION

7.1 Roles and Responsibilities

A resilient community communications network requires defined roles with clear responsibilities. These roles are designed to distribute authority and responsibility so that no single person's absence makes the system fail.

Role	Primary Responsibilities	Minimum Skills Required	Backup Required?
Network Coordinator	Overall system oversight; encryption key management; inter-agency liaison; activation authority; governance leadership	Amateur Technician license minimum; familiarity with all systems; community trust	Yes — designate a Deputy Coordinator
Node Custodians	Maintain specific assigned mesh nodes; regular inspection; battery and solar check; report issues; ensure firmware is current	Comfortable with basic electronics; Meshtastic app familiarity; reliable access to their node	Yes — each node has a primary and backup custodian
Radio Operators	Staff community radio nets; handle formal message traffic; maintain go-box; train others in radio procedures	Amateur Technician license required; General preferred for HF capability	Yes — minimum 3 operators trained for all shifts
Information Runners	Physically carry written messages between nodes; maintain route familiarity; carry runner log sheets; report hazards	Physical fitness; familiarity with runner routes; reliability	Yes — each route has primary and backup runners

Role	Primary Responsibilities	Minimum Skills Required	Backup Required?
Supply Custodian	Maintains USB kits and printed materials; coordinates monthly updates; manages physical supply cache; tracks equipment inventory	Organizational skills; basic computer literacy; reliable storage access	Yes — backup Supply Custodian knows cache location and combination
Training Coordinator	Plans and executes drills and exercises; maintains documentation; onboards new participants; updates manuals and training materials	Familiarity with all community systems; teaching ability; organizational skills	Yes — Assistant Training Coordinator

PRINCIPLE

No single role should be a single point of failure. Every role listed above requires a designated backup. When the primary is unavailable, the backup activates without hesitation. Document both primary and backup contact information in the Emergency Contact Directory (Section 9.1).

7.2 The Neighborhood Cell Structure

A flat, undifferentiated community network becomes unmanageable at scale. The **Neighborhood Cell Structure** organizes the community into manageable geographic units that maintain coordination while preserving decentralization and resilience.

Cell: The smallest unit. Approximately 10–20 households in immediate geographic proximity (a block or small street). Each cell has a **Block Captain** — a community member who knows their neighbors, maintains contact information, and serves as the first point of contact for the cell. The Block Captain is responsible for ensuring every household in the cell has a printed emergency directory, knows the nearest bulletin board, and understands the basic color-flag system.

Cluster: A group of 3–5 cells forms a cluster. Each cluster has at least one mesh network node and at least one licensed radio operator. A **Cluster Coordinator** (typically one of the Block Captains) coordinates between cells and communicates with the broader community network. Clusters cover approximately 50–100 households.

Community Network: Multiple clusters connect to the broader community network via the mesh, radio nets, and the runner system. The Network Coordinator interfaces with the community network, other neighborhoods, and official emergency management.

Benefits of the cell structure:

- Each cell is small enough to manage face-to-face, without any technology
- Loss of any single cell or cluster does not disable the broader network
- Responsibilities are distributed — no central leader is a single point of failure
- The cell structure scales naturally as more households join

7.3 Communication Trees

A **communication tree** (also called a phone tree or radio tree) is a structured system for rapidly distributing a message to all community members. Rather than one person contacting everyone (which is slow and fails if that person is unavailable), each person in the tree contacts 3–5 others, who each contact 3–5 more. A well-built tree can reach 100 people in 4–5 iterations.

Building the Tree

20. The Network Coordinator contacts the 5 Cluster Coordinators.
21. Each Cluster Coordinator contacts the 3–5 Block Captains in their cluster.
22. Each Block Captain contacts the 5–10 household contacts in their cell.
23. Each household contact attempts to contact immediate neighbors without contacts.

Testing the Tree

Test the communication tree at least quarterly. Send a test message (identified as a test) and track: how long it took to reach all participants, which branches failed or

were slow, which contacts could not be reached. Document results and update the tree structure to fix weaknesses found in testing.

TEMPLATE NOTE

A printable Communication Tree template is included in Folder 6 of the USB kit. Fill in contact names and primary/secondary contact methods (radio call sign preferred; telephone secondary) for each position in the tree.

7.4 Decision-Making in Emergencies

Emergencies are not the time for extended deliberation. The community network requires pre-agreed decision-making protocols that allow rapid, authoritative action without requiring consensus — while still remaining accountable to the community.

Pre-agreed activation protocols: Define trigger conditions that automatically activate specific responses. Example: If grid power has been out for more than 2 hours AND the Network Coordinator or Deputy Coordinator cannot be reached by radio within 30 minutes, the senior available Radio Operator activates the radio net and notifies all Cluster Coordinators. No additional authorization is required.

Designated decision-makers: For each category of decision, define who is authorized to decide:

- Mesh network activation: Network Coordinator or Deputy
- Radio net activation: Network Coordinator, Deputy, or senior Radio Operator
- Resource allocation from community cache: Network Coordinator with concurrence of one Cluster Coordinator
- Contact with official emergency management: Network Coordinator only, or Deputy if Coordinator is unreachable

Documentation: Every significant decision made during an emergency must be logged with the time, decision-maker, decision made, and rationale. Use the Incident Log template from the USB kit. This record is essential for After-Action Review and community accountability.

Handling disagreements: During an active emergency, defer to the pre-agreed decision-maker and document the disagreement. Post-emergency, bring unresolved disagreements to the full governance process for discussion and protocol adjustment. Never allow disagreements to paralyze response during an active emergency.

7.5 Mutual Aid Principles

Mutual aid is a framework of reciprocal community support based on solidarity rather than charity. In a mutual aid framework, community members support each other not as donors and recipients, but as equals who recognize that today's resource provider may be tomorrow's person in need.

Core principles for application to community communications:

- **Solidarity, not charity:** This network belongs to the whole community, not to those who built it. Technical expertise does not confer authority over community resources.
- **No gatekeeping:** No community member is excluded from using the communications network based on technical skill, language, income, housing status, or any other factor.
- **Serve the most vulnerable first:** When allocating limited resources — power for charging, runner service, radio time for welfare checks — prioritize: people with medical dependencies, elderly residents who cannot physically travel, residents with disabilities, households with young children, and non-English-speaking residents who may not receive public announcements.
- **Transparent resource distribution:** All community resource decisions (who receives what, when, and how much) are documented and made available to the community. No closed-door resource allocation.
- **No means test:** Do not require community members to prove need before providing access to communications resources. Accept people at their word and verify after the fact when necessary.

7.6 Integration with Official Emergency Management

Community communications networks function alongside — not in opposition to — official emergency management systems. Understanding the official systems and establishing relationships with official agencies before a disaster significantly improves both community and official response effectiveness.

- **CERT (Community Emergency Response Teams):** FEMA-sponsored volunteer training program. CERT-trained community members can assist professional responders and serve as a trained volunteer workforce. Many communities have existing CERT teams — connect with yours and ensure your communications team integrates with their operations plan.
- **Local Emergency Management Agency:** Your county or city emergency management agency coordinates official disaster response. Establish a relationship before an emergency. Provide them with your network's contact information and capabilities. Ask about their communications plan and how community networks can support it.
- **ARES/RACES (Amateur Radio Emergency Service / Radio Amateur Civil Emergency Service):** ARES is the ARRL's emergency communications volunteer program. RACES is the FCC-authorized emergency radio service for civil defense. Encourage your licensed amateur operators to affiliate with local ARES and RACES groups. These organizations provide training, coordination, and official recognition for amateur radio emergency communications.
- **American Red Cross:** The Red Cross coordinates shelter, mass feeding, and disaster relief operations. Establish communication links with your local chapter's emergency communications plan.

When to defer to official systems: When official emergency services are active and operational, defer to their command structure for public safety decisions. The community network's role is to support and supplement official response — providing neighborhood-level situational awareness, welfare checks on residents, and resource distribution information that official systems often lack the capacity to gather. Never interfere with official radio frequencies or communications channels.

SECTION 8

DRILLS AND EXERCISES

8.1 Why Regular Drills Matter

Skills that are not practiced degrade. A radio operator who passed their license exam three years ago and has not transmitted since will struggle to run a net under pressure. A community that deployed its mesh network once at a picnic and never tested it in a simulated emergency will find critical failures when a real emergency occurs. Drills exist to surface these failures in a controlled environment — when failure is an opportunity to learn rather than a threat to lives.

Beyond technical skill maintenance, regular drills build three things that cannot be acquired any other way: **muscle memory** (procedures executed without conscious effort), **trust** (community members who have worked together under simulated stress know they can rely on each other under real stress), and **confidence** (participants who have successfully executed their roles in exercises approach real emergencies with confidence rather than panic).

The single most common failure mode in community emergency preparedness is this: the group builds a system, declares it ready, and never tests it. When a real emergency arrives, the batteries are dead, the firmware is outdated, nobody remembers the encryption key, and the network coordinator has moved away. **Test everything. Test it regularly. Fix what fails.**

8.2 Exercise Types

Exercise Type	Duration	Participants	What It Tests	Recommended Frequency
Communications Check (Radio Net)	15–30 minutes	Radio operators and check-in participants	Radio operation, call signs, net control procedure, equipment function	Weekly or monthly
Tabletop	2–4 hours	All role holders;	Decision-	Monthly

Exercise Type	Duration	Participants	What It Tests	Recommended Frequency
Exercise		leadership	making, protocols, inter-role coordination, scenario knowledge	
Partial Activation	Half-day (4–6 hours)	Specific subsystem (mesh only, or radio only)	Full deployment and operation of one system pillar; equipment readiness; role fulfillment	Quarterly
Full Activation	Full day (8–12 hours)	All participants; whole community invited	All three pillars simultaneously; community integration; inter-cluster coordination; realistic scenarios	Annually
Surprise Drill	Variable (1–4 hours)	On-call role holders (no advance notice)	Real-world readiness; response without preparation; equipment actually ready status	Ad hoc, 1–2 times per year

8.3 Monthly Radio Net Procedure

The monthly (or weekly) radio net is the foundation of ongoing operator readiness. It verifies that equipment works, operators remember procedure, and the community has a regular point of contact to share updates.

Sample Net Control Script

NET CONTROL SCRIPT — [COMMUNITY NAME] EMERGENCY COMMUNICATIONS NET

Net Control reads the following, substituting bracketed information:

"This is [CALL SIGN], [CALL SIGN], [CALL SIGN]. I am opening the [COMMUNITY

NAME] Emergency Communications Net for [DATE]. This is a [scheduled / unscheduled] net. All stations please stand by."

[Pause 10 seconds]

"Stations with priority traffic, please identify now."

[Handle any priority traffic before continuing.]

"The [COMMUNITY NAME] net is now open for check-ins. Please identify with your call sign and location. All stations, check in now."

[Record all check-ins on the Net Log Sheet.]

"[NUMBER] stations have checked in. Is there any traffic for the net?"

[Handle traffic in order of priority. Pass each piece of traffic, confirm receipt.]

"Any announcements for the net?"

[Allow brief announcements.]

"This concludes the [COMMUNITY NAME] Emergency Communications Net. [NUMBER] stations checked in. [NUMBER] pieces of traffic handled. This is [CALL SIGN] closing the net at [TIME]. Good [morning/afternoon/evening] all."

Check-In Procedure for Participants

24. Listen to ensure the net is in check-in phase before transmitting.
25. When Net Control opens check-ins, transmit your call sign and location: "[CALL SIGN], [LOCATION], [any traffic]."
26. Wait for Net Control to acknowledge you.
27. If you have traffic (a message to pass), indicate this during check-in: "[CALL SIGN], [LOCATION], one piece of traffic for [recipient]."

8.4 Mesh Network Stress Test

The mesh network stress test verifies that the network performs as expected under realistic conditions, including node failures.

Stress Test Procedure

28. **Baseline test:** With all nodes active, send a message from the most distant node to the hub node. Confirm delivery time and route. Record baseline RSSI (signal strength) values between all node pairs.
29. **Single-node failure simulation:** Power down one non-hub node. Confirm that messages from beyond the removed node still reach the hub via alternative routes. Measure delivery time impact.
30. **Multi-node failure simulation:** Remove two or more nodes simultaneously. Identify which combinations cause network partitioning (isolation). Use results to identify coverage gaps requiring additional nodes.
31. **Load test:** Have all community Meshtastic handhelds send simultaneous messages. Observe channel congestion. Identify the maximum number of simultaneous users before message delivery degrades unacceptably.
32. **Non-technical user test:** Ask community members who are not part of the technical team to use the Meshtastic app on their smartphones to send a message and confirm receipt. Identify usability barriers. Simplify if needed.
33. **Coverage walk:** Walk the community coverage area with a handheld Meshtastic device, sending messages at 100-meter intervals. Map signal quality to identify dead zones. Update the coverage map.

8.5 After-Action Reviews

Every exercise — and every real activation — should be followed by a structured After-Action Review (AAR) within 48 hours while memories are fresh. The AAR is not a blame session; it is a learning process. All participants should feel safe to identify failures honestly, knowing the goal is improvement, not accountability.

AAR Template

Section	Questions to Address	Notes
What Was Planned	What were the exercise objectives? What was supposed to happen?	

Section	Questions to Address	Notes
What Happened	What actually occurred? Where did the exercise deviate from the plan?	
What Went Well	Which systems, procedures, and individuals performed as expected or better?	
What Needs Improvement	Where did the system fail or underperform? What surprised you? What was hard?	
Action Items	What specific changes will be made? Who is responsible? By what date?	

Action Item Tracking

#	Action Item	Owner	Due Date	Status
1				
2				
3				
4				
5				

8.6 Annual Full-System Exercise

Once per year, the community should conduct a full-day exercise that activates all three communications pillars simultaneously, involves as much of the community as possible, and uses realistic scenarios to test the complete system.

Exercise Planning Timeline

- **8 weeks before:** Select date; assign exercise planning team; develop scenario
- **6 weeks before:** Distribute participant notices; recruit evaluators; finalize scenario
- **4 weeks before:** Conduct pre-exercise briefing for all role holders; verify equipment status

- **2 weeks before:** Tabletop run-through of scenario; confirm logistics; prepare inject schedule
- **Exercise Day:** Brief all participants; run the exercise; evaluators observe and document
- **Within 48 hours:** Conduct AAR; document findings; assign action items

Sample Exercise Scenario Injects (Schedule)

Time	Inject	Systems Tested
0800	EXERCISE BEGIN: Simulated power grid failure. All grid power declared unavailable.	Battery backup, solar activation, equipment transition
0815	Network Coordinator "unavailable" — Deputy Coordinator must take over activation	Backup roles, activation protocols
0830	Radio net opened; all clusters check in; welfare check of vulnerable residents begins	Radio net, communication tree, cell structure
0900	Inject: Node NORTHSIDE-ROOF-01 has failed. Coverage gap in Northside cluster.	Mesh resilience, alternative routing, runner deployment
1000	Inject: Report of household needing medical supplies — resource allocation required	Mutual aid protocols, runner system, decision-making
1100	Inject: Unverified report of road closure on main runner route	Information verification, alternate routes, bulletin board update
1200	Lunch break / resource consolidation exercise	Power conservation, supply management
1300	Official Emergency Management contact — coordinate resource request	Inter-agency communication protocols, HF radio if available
1400	Inject: Community member challenges resource allocation decision — governance test	Decision-making protocols, documentation, community governance
1500	EXERCISE END. Stand-down. All systems secure.	Full system
1530	After-Action Review begins	All

SECTION 9

QUICK REFERENCE AND RESOURCES

9.1 Emergency Contact Directory (Fill-in Template)

Role	Name	Call Sign	Primary Contact	Secondary Contact	Notes
Network Coordinator					
Deputy Coordinator					
Node Custodian 1					Node: _____
Node Custodian 2					Node: _____
Node Custodian 3					Node: _____
Node Custodian 4					Node: _____
Node Custodian 5					Node: _____
Radio Operator 1					
Radio Operator 2					
Radio Operator 3					
Radio Operator 4					
Radio					

Role	Name	Call Sign	Primary Contact	Secondary Contact	Notes
Operator 5					
Information Runner 1					Route: _____
Information Runner 2					Route: _____
Information Runner 3					Route: _____
Information Runner 4					Route: _____
Information Runner 5					Route: _____
Supply Custodian					
Training Coordinator					
Local Emergency Mgmt					
Local Amateur Radio Club					
Local ARES/RACES EC					
CERT Team Leader					

9.2 Node Registry Template

Node ID	Physical Location	Custodian	Hardware	Installed Date	Last Check	Status

Node ID	Physical Location	Custodian	Hardware	Installed Date	Last Check	Status

9.3 Frequency Quick Reference

Frequency	Service	Purpose	License
162.400–162.550 MHz	NOAA Weather Radio	Weather alerts — monitor continuously	None (RX)
146.520 MHz	2m Amateur	National calling frequency — make contact, then QSY	Technician+
146.580 MHz	2m Amateur	National simplex emergency (backup)	Technician+
462.675 MHz	GMRS Ch. 20	GMRS calling/emergency	GMRS
462.5625 MHz	FRS/GMRS Ch. 1	FRS short-range coordination	None (FRS power)
27.065 MHz (Ch. 9)	CB Radio	CB emergency/distress channel	None
144.390 MHz	2m APRS	APRS position and messaging	Technician+
915 MHz	LoRa ISM / Meshtastic	Community mesh network primary	None (Part 15)
3.985 MHz	80m HF Amateur	NTS traffic net (evening)	General+
7.285 MHz	40m HF Amateur	NTS traffic net (daytime)	General+
14.300 MHz	20m HF Amateur	Maritime Mobile — worldwide emergency	General+
7.177 MHz	40m HF Amateur	Winlink/Vara HF	General+

Frequency	Service	Purpose	License
		(varies by region)	
Local Repeater: _____		Primary community repeater	
Backup Repeater: _____		Backup community repeater	

9.4 Glossary of Terms

APRS — Automatic Packet Reporting System. A digital protocol operating on 144.390 MHz (North America) that transmits position, weather, and messaging data via packet radio. Does not require internet access.

BATMAN-adv — Better Approach To Mobile Adhoc Networking — advanced. A Linux kernel module that enables WiFi mesh networking at layer 2 of the network stack. Used in community WiFi mesh projects.

Briar — An open-source Android messaging app that communicates via Bluetooth, WiFi Direct, or the Tor network without requiring internet access. Contacts must be added in person by scanning a QR code.

Call Sign — An FCC-assigned alphanumeric identifier (e.g., W8XYZ) used to identify a licensed radio station or operator. Must be stated at the beginning and end of each transmission.

Channel — In radio: a specific frequency or frequency pair designated for a particular use. In Meshtastic: a named communication channel with an associated pre-shared encryption key.

Dead Drop — A pre-agreed physical location where messages can be left and retrieved without direct contact between sender and recipient. Enables asynchronous physical communication.

DMR — Digital Mobile Radio. An ETSI digital voice and data radio standard widely used in commercial, GMRS, and amateur radio applications. Supports two simultaneous time-division channels on one frequency.

EMCOMM — Emergency Communications. The discipline and practice of maintaining radio communications capability for use during disasters and emergencies. Associated with ARES, RACES, and CERT programs.

FCC — Federal Communications Commission. The US federal agency that regulates radio communications, issues licenses, and enforces the Communications Act.

FRS — Family Radio Service. An unlicensed UHF radio service operating on 22 channels between 462 and 467 MHz. Limited to 2W on most channels. Widely available in consumer electronics stores.

GMRS — General Mobile Radio Service. A licensed UHF radio service (FCC license, $35, no exam) with higher power limits than FRS and access to repeater operation. The GMRS license covers the licensee and immediate family members.

Go-Box — A portable, self-contained emergency communications station assembled in a weatherproof case. Contains a transceiver, power supply, antenna, cables, and operating supplies. Can be deployed and operational in under 15 minutes. See Appendix D.

HF — High Frequency. The radio frequency range of 3–30 MHz. HF signals can travel thousands of miles by reflecting off the ionosphere. Requires Amateur General or Extra class license for most HF amateur bands.

J-Pole — A half-wave antenna fed by a quarter-wave matching stub. Provides modest gain over a simple ground-plane antenna with no ground radials required. Well-suited for VHF and UHF mesh/radio node installations.

JS8Call — A weak-signal messaging application for HF radio derived from WSJT-X/FT8. Enables keyboard-to-keyboard messaging, relaying, and store-and-forward networking over HF frequencies.

Kiwix — An open-source offline content reader available for Android, iOS, Windows, Mac, and Linux. Reads ZIM format files. Used to access Wikipedia and other reference content without internet access.

LoRa — Long Range. A spread-spectrum radio modulation technique optimized for long-range, low-power, low-data-rate communication. Used as the underlying radio technology in Meshtastic nodes. Typical range: 1–10+ km per hop.

Meshtastic — An open-source firmware and application ecosystem that implements a mesh network over LoRa radio hardware. Provides text messaging, GPS position sharing, and store-and-forward capability without internet access.

Mesh Network — A network topology in which every node connects to multiple other nodes, creating multiple paths for data to travel. The network self-heals around failed nodes by routing through alternative paths.

Net Control — The designated operator who manages a radio net. Net Control calls the net to order, takes check-ins, manages traffic, and closes the net. Net Control is not a permanent authority role — it rotates among trained operators.

Node — In a mesh network: a single device that participates in the mesh as both a client and a relay. In community context: a fixed, solar-powered Meshtastic device mounted at an elevated location to extend network coverage.

OPSEC — Operational Security. The practice of protecting sensitive information from disclosure to unauthorized parties. In community communications: avoid transmitting personal details, locations of supplies, or security-sensitive information over unencrypted or public channels.

Phonetic Alphabet — A system of words representing letters of the alphabet used to spell out words clearly over radio. See Appendix A for the NATO standard alphabet used in emergency communications.

PSK31 — Phase Shift Keying 31 baud. A narrow-bandwidth HF digital keyboard-to-keyboard communication mode. Operates in very limited spectrum and is useful for direct one-on-one HF text communication.

QRM — Amateur radio shorthand for interference from other radio signals. Q-codes (Q-signals) are a standardized system of abbreviations used in radio communication.

QSO — Amateur radio shorthand for a radio contact or conversation between two stations.

Raspberry Pi — A low-cost, credit-card-sized Linux computer widely used in electronics projects. In community mesh networks: used as a gateway/hub node, running Meshtastic Python integration, store-and-forward messaging server, or BATMAN-adv mesh router.

Repeater — An automated radio station that receives signals on one frequency (input) and simultaneously retransmits them at higher power on another frequency (output). Greatly extends the range of portable and mobile radio equipment.

Reticulum — An open-source cryptographic networking stack designed to operate over low-bandwidth radio links, serial connections, and other constrained media. Provides strong encryption and resilient routing. Used with Nomad Network and LXMF applications.

SALUTE — A military-derived format for reporting emergency information. Stands for: Size, Activity, Location, Unit/Uniform, Time, Equipment. See Appendix B.

SDR — Software Defined Radio. A radio receiver (and sometimes transmitter) in which the radio hardware is simplified and most signal processing is done in software. The RTL-SDR dongle ($25–$35) is a popular receive-only SDR for monitoring a wide range of signals.

Simplex — Radio communication on a single frequency, used for both transmitting and receiving. Unlike repeater operation (duplex), simplex is direct station-to-station. The national 2-meter simplex calling frequency is 146.520 MHz.

Solar Charge Controller — A device that regulates the voltage and current flowing from a solar panel to a battery, preventing overcharging and battery damage. Two types: PWM (simpler, less efficient) and MPPT (more efficient, recommended for larger systems).

Store-and-Forward — A messaging pattern in which a message is stored at an intermediate node when the destination is not reachable, and forwarded when a connection is established. Used in Meshtastic and Winlink to compensate for intermittent connectivity.

UHF — Ultra High Frequency. The radio frequency range of 300–3,000 MHz. UHF signals travel by line-of-sight but penetrate buildings better than VHF. FRS, GMRS, and the 70-centimeter amateur band are in the UHF range.

USB Kit — The Community USB Kit: a standardized USB flash drive (32 GB minimum) containing all community communications documentation, software, maps, and reference data. Distributed to all key role holders and updated monthly.

VHF — Very High Frequency. The radio frequency range of 30–300 MHz. VHF signals travel primarily by line-of-sight, with range enhanced by antenna height. NOAA Weather Radio, 2-meter amateur band, and FM broadcast radio are in the VHF range.

Winlink — A global radio email system that allows sending and receiving email over HF, VHF, and UHF radio links without internet access. Uses Pactor, VARA,

ARDOP, or packet protocols. Widely used by emergency communicators for formal message traffic.

Yagi — A directional antenna consisting of a driven element and multiple parasitic elements (directors and reflectors) along a boom. Provides significant gain in one direction. Used for point-to-point links between mesh nodes or for working distant repeaters.

ZIM File — The file format used by the Kiwix offline reader. Wikipedia and other large reference sites are available as ZIM files for download and offline use. The English Wikipedia ZIM file is approximately 85 GB (full) or 6 GB (simplified).

9.5 Recommended Reading and Resources

Resource	Type	Offline Available?	Notes
ARRL Handbook for Radio Communications	Reference book	Yes (print or PDF)	The definitive amateur radio technical reference. Updated annually. Covers all modes, antennas, propagation, and emergency operations.
ARRL Emergency Communications Handbook	Reference book	Yes (print or PDF)	EMCOMM-specific guidance for amateur radio operators. Net operations, ICS integration, message handling.
Meshtastic Documentation	Online documentation / PDF	Yes (PDF download)	Official Meshtastic project documentation. Device configuration, firmware, Python API. Include in USB kit.
FEMA IS-100, IS-200, IS-700	Online course / PDF	Yes (PDF print)	Free FEMA independent study courses on ICS, NIMS, and emergency management. Foundational for working with official

Resource	Type	Offline Available?	Notes
			systems.
RepeaterBook	Online directory / CSV export	Yes (CSV download)	Comprehensive repeater directory. Download CSV for your region and store in USB kit DATA folder.
Winlink.org Documentation	Online / PDF	Yes (PDF)	Comprehensive guides for Winlink email-over-radio. Include VARA HF/FM guides.
Wikipedia (Offline via Kiwix)	Encyclopedia / ZIM file	Yes	Offline access to Wikipedia's entire content base. Invaluable for medical, technical, and reference use during extended outages.
JS8Call Documentation	Online / PDF	Yes (PDF)	Documentation for JS8Call HF weak-signal messaging. Operating guide and API reference.
The Art of Community by Jono Bacon	Book	Yes (free PDF)	Practical community building guidance applicable to organizing a volunteer communications network. Free download from author.

9.6 Version History and Update Log

Version	Date	Changes	Updated By
1.0	2026	Initial release — First Edition. All 9 sections and 4 appendices.	Community Network Team

Version	Date	Changes	Updated By

APPENDIX A — NATO PHONETIC ALPHABET

Use the NATO Phonetic Alphabet to spell letters clearly over radio when there is any chance of misunderstanding. Always use this alphabet for call signs, addresses, and critical identifiers.

Letter	Word	Pronunciation	Letter	Word	Pronunciation
A	**Alfa**	**AL-fah**	**N**	**November**	**no-VEM-ber**
B	**Bravo**	**BRAH-voh**	**O**	**Oscar**	**OSS-cah**
C	**Charlie**	**CHAR-lee**	**P**	**Papa**	**pah-PAH**
D	**Delta**	**DELL-tah**	**Q**	**Quebec**	**keh-BECK**
E	**Echo**	**ECK-oh**	**R**	**Romeo**	**ROW-me-oh**
F	**Foxtrot**	**FOKS-trot**	**S**	**Sierra**	**see-AIR-rah**
G	**Golf**	**GOLF**	**T**	**Tango**	**TANG-go**
H	**Hotel**	**hoh-TELL**	**U**	**Uniform**	**YOU-nee-form**
I	**India**	**IN-dee-ah**	**V**	**Victor**	**VIK-tah**
J	**Juliett**	**JEW-lee-ett**	**W**	**Whiskey**	**WISS-key**
K	**Kilo**	**KEY-loh**	**X**	**X-ray**	**ECKS-ray**
L	**Lima**	**LEE-mah**	**Y**	**Yankee**	**YANG-key**
M	**Mike**	**MIKE**	**Z**	**Zulu**	**ZOO-loo**

Numeric Pronunciations

Digit	Pronunciation	Digit	Pronunciation
0	ZEE-ro	**5**	FIF-tuh
1	WUN	**6**	SIX
2	TOO	**7**	SEV-en

Digit	Pronunciation	Digit	Pronunciation
3	TREE	**8**	AIT
4	FOW-er	**9**	NIN-er

APPENDIX B — SALUTE REPORT FORMAT

The SALUTE format is a structured method for reporting emergency observations clearly and completely over radio. Using SALUTE ensures the receiver gets all critical information without needing to ask follow-up questions. Memorize this format — in a high-stress situation, having a checklist prevents critical information from being omitted.

Letter	Element	What to Report	Example
S	Size	How many people, vehicles, or units are involved? Estimate if exact count is unknown.	"Approximately 12 people"
A	Activity	What is happening? Be specific: injured, trapped, evacuating, requesting supplies, fire, etc.	"Three individuals appear injured and unable to walk"
L	Location	Exact address, cross streets, or landmark. Include direction if relevant. Provide GPS coordinates if available.	"Corner of Maple Ave and 5th Street, northeast corner"
U	Unit / Uniform	Who is involved? Civilians, children, elderly, first responders? Are they identifiable by clothing, markings, or identifying features?	"All civilians, mix of adults and children, no emergency personnel on scene"
T	Time	When did you observe this? Or: how long has the situation been ongoing? Use 24-hour time where	"Observed at 1435 hours, approximately 20 minutes ago"

Letter	Element	What to Report	Example
		possible.	
E	Equipment	What resources are on scene or needed? Include medical supplies, vehicles, personnel, water, structural damage, fire, etc.	"Need: first aid, stretcher, at least two additional people for assistance. On scene: none."

SALUTE Fill-In Report Form

SALUTE REPORT — Copy and use for each incident	
Reporting Party:	
Call Sign:	
Time of Report:	
S — Size:	
A — Activity:	
L — Location:	
U — Unit/Uniform:	
T — Time Observed:	
E — Equipment (On Scene / Needed):	
Received By:	
Action Taken:	

APPENDIX C — MESH NODE ASSEMBLY CHECKLIST

Use this checklist when assembling and deploying each new community mesh node. Check off each item as completed. Do not skip steps. Sign and date the completed checklist and file it with the Node Registry.

Phase 1 — Procurement and Preparation

- LoRa board received and verified (model: _____)

- Antenna received and inspected for damage (type: _____)
- SMA pigtail cable received (length: _____)
- Solar panel received and tested (watts: _____)
- Battery received and charge-tested (Ah: _____, chemistry: _____)
- MPPT charge controller received
- 5V step-down converter received and tested
- Weatherproof enclosure received and inspected
- SMA bulkhead connector and cable gland received
- Mounting hardware assembled (mast clamp, U-bolts, screws)
- All items logged in inventory

Phase 2 — Firmware and Configuration

- Downloaded correct Meshtastic firmware for board model
- Flashed firmware via USB using Meshtastic Web Flasher (firmware version: _____)
- Verified device appears in Meshtastic app via Bluetooth
- Set device name per community naming convention: _____
- Set region to US (915 MHz)
- Configured community channel name and pre-shared key (PSK)
- Set GPS (if equipped) or entered manual coordinates: Lat _____ / Lon _____
- Verified device name and GPS position visible on community mesh
- Set hop limit (recommended: 3 for standard nodes, 7 for router nodes)
- Set device role (CLIENT, ROUTER, or ROUTER_CLIENT as appropriate)

Phase 3 — Hardware Assembly

- Drilled enclosure for cable gland and bulkhead connector

- Installed SMA bulkhead connector through enclosure wall (weatherproof)
- Installed cable gland for power cable
- Connected LoRa board to SMA pigtail to bulkhead connector
- Mounted LoRa board inside enclosure (secure, not loose)
- Connected step-down converter output to board USB power input
- Wired charge controller: solar panel input, battery output, load output
- Connected battery to charge controller (correct polarity verified)
- Connected solar panel to charge controller (correct polarity verified)
- Verified charge controller shows correct charging status
- Verified board powers on from battery
- Sealed all cable entry points with silicone sealant
- Closed and latched weatherproof enclosure

Phase 4 — Pre-Deployment Testing

- Pre-mounting radio link test passed (test with nearest existing node)
- Message delivery confirmed: sent test message, received on community mesh
- GPS fix confirmed (if equipped)
- Battery voltage verified (above 12.4V for lead-acid, above 13V for LiFePO4)

Phase 5 — Mounting and Deployment

- Antenna mounted at highest practical point and secured
- Antenna oriented vertically (unless directional Yagi)
- Coax connections weatherproofed with self-amalgamating tape
- Enclosure mounted securely (no movement in wind test)
- Solar panel mounted at correct angle for latitude (facing south)
- All cables secured and protected from abrasion and UV exposure
- Post-mounting range test completed

- Coverage map updated
- Node Registry entry completed and filed

Phase 6 — Handoff

- Node Custodian briefed and has physical access to node
- Custodian has been shown how to check battery voltage and solar charging status
- Custodian has Network Coordinator contact information
- Backup Custodian assigned and aware of node location
- Node ID: _____ | Installation Date: _____ | Custodian: _____
- Completed checklist signed by installer and custodian and filed with Network Coordinator

Installer Signature: _____ Date: _____

Custodian Signature: _____ Date: _____

APPENDIX D — GO-BOX BUILD GUIDE

A **go-box** is a portable, self-contained emergency communications station assembled in a weatherproof, carry-ready case. The go-box concept comes from military and professional emergency communications practice: when you need to establish communications in an unfamiliar location with no infrastructure, you carry everything you need in one container. A well-built go-box can be deployed and fully operational in under 15 minutes.

The community should maintain at least one go-box, stored at the Network Coordinator's location or a designated community supply point, charged and ready at all times.

Case Selection

The case must be waterproof, impact-resistant, and carry-ready. Options:

- **Pelican 1510 or 1550:** Airline carry-on size (1510) or larger. Military-grade waterproof. Excellent choice for a full HF/VHF station. Cost: $150–$350.
- **Apache 4800 (Harbor Freight):** Budget-friendly waterproof case. Similar internal volume to Pelican 1510. Cost: $30–$60.
- **Commercial ammo can or military surplus case:** Inexpensive and durable. May require additional weatherproofing.

Go-Box Component List

Component	Specification / Model	Approximate Cost	Purpose
Carrying Case	Pelican 1510 or Apache 4800	$40–$200	Weatherproof housing for all components
VHF/UHF Transceiver	Yaesu FT-60R or FT-3DR, or mobile unit	$100–$400	Primary voice communications
HF Transceiver (optional/advanced)	Xiegu G90, IC-705, or portable HF rig	$400–$1,200	Long-distance HF communications
Battery (LiFePO4)	12V, 20–40 Ah LiFePO4	$80–$200	Primary power source for all equipment
Solar Panel (foldable)	40–100W foldable panel	$60–$150	Battery recharging in the field
MPPT Charge Controller	10A MPPT controller	$25–$45	Solar-to-battery charge management
VHF/UHF Antenna + Mount	Portable J-pole or magnetic mount whip	$20–$60	Deployable antenna for handheld or mobile radio
HF Antenna (optional)	End-fed half-wave (EFHW) wire antenna, 40m–10m	$50–$150	HF communications; deployable with fishing pole mast or thrown into tree
Antenna Mast (collapsible)	Fiberglass push-up mast (10–20 ft) or telescoping	$40–$120	Elevates antenna for improved range
Meshtastic Node (handheld)	TTGO T-Beam or Heltec LoRa 32	$35–$55	Mesh network integration from go-box location
Laptop or Tablet	Any Windows laptop with CHIRP, Winlink,	Existing equipment	Digital modes (Winlink, JS8Call),

Component	Specification / Model	Approximate Cost	Purpose
	JS8Call installed		radio programming, USB kit access
Soundcard Interface	SignaLink USB or Digirig Mobile	$30–$110	Connects radio to laptop for digital modes (Winlink, JS8Call, etc.)
USB Kit Drive	32 GB USB, current contents	$10	All community data, maps, software, documentation
Power Distribution Panel	Anderson Powerpole fused distribution block	$20–$50	Clean, fused power distribution to all equipment
Cables and Adapters	Coax, Anderson Powerpole cables, USB cables, audio patch cables	$30–$80	Interconnects for all components
Operating Supplies	Message log pads, pencils, pens, operator's manual, laminated quick-cards	$10–$20	Paper-based backup; reference materials
Headset or Headphones	Noise-canceling headset with PTT	$20–$80	Comfortable extended operation; noise isolation
LED Lantern (battery)	USB rechargeable LED lantern	$15–$30	Lighting for nighttime operation

Assembly Instructions

34. **Plan the layout.** Before drilling or mounting anything, arrange all components in the case and confirm everything fits with adequate clearance for ventilation and cable routing.
35. **Install power distribution.** Mount the Anderson Powerpole distribution block in an accessible corner. Run the main positive and negative leads from the battery to the fused distribution block. Fuse each branch circuit individually (transceiver: 20–30A; Meshtastic node: 1A; charge controller: per panel rating).

36. **Mount equipment.** Use hook-and-loop (Velcro) tape, custom foam inserts, or 3D-printed brackets to secure all components. Nothing should slide or rattle. Foam padding protects delicate equipment from impact.
37. **Route cables cleanly.** Use cable ties and cable guides to route all cables neatly. Label every cable at both ends. No cable should be able to pull loose under normal use.
38. **Label everything.** Label all controls, connectors, switches, and fuses. A field operator who did not build the box must be able to operate it without a manual.
39. **Load the USB kit.** Place the current USB kit in a designated compartment or cable pocket. Label it clearly. Verify it contains current data before each deployment.
40. **Document the configuration.** Create a one-page "Go-Box Quick Start Guide" listing: power-up sequence, channel settings, pre-programmed frequencies, and emergency contact frequencies. Laminate it and attach to the inside lid.

Maintenance Schedule

Frequency	Maintenance Task
Monthly	Power on all equipment; verify function; update USB kit with current data; check battery voltage; verify solar panel connection
Quarterly	Full deployment exercise; check all cable connections; update radio programming files; verify antenna condition; test digital mode operation (Winlink, JS8Call)
Annually	Full systems check; firmware updates for all devices; battery capacity test; replace any worn cables or components; update Quick Start Guide if configuration has changed
After each deployment	Recharge battery fully; note any issues in the maintenance log; replace any used consumables; report any equipment failures to Network Coordinator

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> "This manual is released for free community use.
>
> Reproduce freely. Distribute widely.
>
> When infrastructure fails, community survives."

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